

TRAJECTORY SMOOTHNESS: C^0 STEP JUMP VS C^2 QUINTIC CONTINUITY

Bounding Jerk $j(t) = d^3q/dt^3$ to Prevent Joint Mechanical Shock and Resonant Vibration

× C^0 ZERO-ORDER HOLD (Step Jumps)

Direct execution of raw neural waypoints



- Infinite theoretical acceleration at boundaries
- Gearbox tooth backlash impact & wear
- Excites high-frequency structural resonances
- **Unacceptable in physical robots**

✓ C^2 QUINTIC POLYNOMIAL SPLINE

Continuous position, velocity, and acceleration



- Zero velocity or acceleration discontinuities
- Smooth motor current rise within inverter slew limits
- Clean torque tracking with minimum heat generation
- **Standard for physical deployment**